

# Online Appendix to “Lookout 2: Anomaly Detection via Leave-One-Out Kernel Density Estimation”

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## S1 Notation and results

Sections S1 to S3 of this appendix prove Proposition 1 and Counterexample 1 of the main paper, restated below; Section S4 reports additional experiments. Throughout,  $\mathbf{y}_1, \dots, \mathbf{y}_n$  are sampled independently from a density  $f$  on  $\mathbb{R}^m$ ,  $T$  is their Euclidean minimum spanning tree with ordered edge lengths  $e_{(1)} \leq \dots \leq e_{(n-1)}$ , and  $h_n = e_{(\lceil \gamma(n-1) \rceil)}$  for a fixed  $\gamma \in (0, 1)$ . For  $r > 0$ ,  $C_n(r)$  is the number of connected components of the graph that joins every pair of observations at distance at most  $r$ . The edges of  $T$  of length at most  $r$  form a spanning forest of that graph, so there are  $n - C_n(r)$  of them, and  $e_{(k)} \leq r$  exactly when  $C_n(r) \leq n - k$ . The ordered nearest-neighbour distances are  $D_{(1)} \leq \dots \leq D_{(n)}$ . For  $\lambda > 0$ ,  $\mathcal{P}_\lambda$  is a homogeneous Poisson process of intensity  $\lambda$  on

$\mathbb{R}^m$ ; joining each pair of points of  $\mathcal{P}_\lambda \cup \{\mathbf{0}\}$  at distance at most 1,  $\mathcal{C}_\lambda$  is the component containing the origin, and  $\kappa(\lambda) = \mathbb{E}[1/|\mathcal{C}_\lambda|]$ , with  $1/|\mathcal{C}_\lambda| = 0$  when the component is infinite. Finally,  $b_m$  denotes the volume of the unit ball in  $\mathbb{R}^m$ ,  $s_{m-1} = mb_m$  the  $(m-1)$ -dimensional surface measure of the unit sphere  $S^{m-1} \subset \mathbb{R}^m$ , and  $B(\mathbf{x}, r)$  the closed ball of radius  $r$  centred at  $\mathbf{x}$ .

**Proposition 1.** *Let  $\mathbf{y}_1, \dots, \mathbf{y}_n$  be sampled independently from a density  $f$  on  $\mathbb{R}^m$ , and fix  $\gamma \in (0, 1)$ . Then  $n^{1/m}h_n \rightarrow q_\gamma(f)$  in probability as  $n \rightarrow \infty$ , where  $q_\gamma(f) \in (0, \infty)$  is the unique solution  $q$  of*

$$\int_{\mathbb{R}^m} \kappa(q^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x} = 1 - \gamma. \quad (\text{S1})$$

Moreover  $t_- \leq q_\gamma(f) \leq t_+$ , where  $t_-$  and  $t_+$  are the unique positive solutions  $t$  of

$$\int_{\mathbb{R}^m} \exp(-b_m t^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x} = 1 - \gamma \quad \text{and} \quad \int_{\mathbb{R}^m} (1 - \exp(-b_m t^m f(\mathbf{x}))) d\mathbf{x} = (1 - \gamma) b_m t^m$$

respectively, and  $b_m$  is the volume of the unit ball in  $\mathbb{R}^m$ .

**Counterexample 1.** *Let the observations be sampled independently from the density  $f(\mathbf{x}) = c_m/(1 + \|\mathbf{x}\|^{m+1})$  on  $\mathbb{R}^m$ ,  $m \geq 1$ , and put  $\omega_n = n - \lfloor n^{1/4}/4 \rfloor$ , which is at most  $n-1$  once  $n \geq 256$ . Then  $\mathbb{P}(e_{(\omega_n)} > n^{1/8}) \rightarrow 1$  as  $n \rightarrow \infty$ . In particular  $e_{(\omega_n)}$ , and every longer edge of  $T$ , tends to infinity in probability.*

## S2 Proof of Proposition 1

Write  $\mathcal{Y}_n = \{\mathbf{y}_1, \dots, \mathbf{y}_n\}$  and  $k_n = \lceil \gamma(n-1) \rceil$ .

*Proof of Proposition 1. Step 1: the component count.* For  $t > 0$  put  $\Xi_n(t) := n^{-1}C_n(tn^{-1/m})$  and

$$G(t) := \int_{\mathbb{R}^m} \kappa(t^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x}.$$

Joining at distance  $r_n = tn^{-1/m}$  keeps  $nr_n^m = t^m$  fixed, which is the thermodynamic regime, so the law of large numbers for component counts, Mitsche & Penrose (2021, Theorem 3.13, an application of their Theorem 2.1), gives  $\Xi_n(t) \rightarrow G(t)$  in probability, for every  $t > 0$  and every density  $f$ . Their

limit function for component counts is  $\kappa$ , and their convergence is in fact complete and in  $L^2$ . The same conclusion follows from the law of large numbers for the empirical distribution of the rescaled edge lengths of  $T$  (Penrose & Yukich 2003, Theorem 2.3(i), first obtained by Penrose 1996), since  $n - C_n(r)$  is the number of edges of  $T$  of length at most  $r$ ; the component-count form is used here because its limit is expressed through  $\kappa$ .

*Step 2: properties of  $G$ .* The origin forms a component on its own exactly when  $\mathcal{P}_\lambda$  has no point in  $B(\mathbf{0}, 1)$ , so  $\kappa(\lambda) \geq \mathbb{P}(|\mathcal{C}_\lambda| = 1) = \exp(-b_m \lambda)$ . Its component contains the  $N \sim \text{Poisson}(b_m \lambda)$  points of  $\mathcal{P}_\lambda$  in  $B(\mathbf{0}, 1)$ , so  $\kappa(\lambda) \leq \mathbb{E}[1/(1 + N)] = (1 - \exp(-b_m \lambda))/(b_m \lambda)$ . Integrating,

$$G_-(t) := \int_{\mathbb{R}^m} \exp(-b_m t^m f(\mathbf{x})) f(\mathbf{x}) d\mathbf{x} \leq G(t) \leq G_+(t) := \frac{1}{b_m t^m} \int_{\mathbb{R}^m} (1 - \exp(-b_m t^m f(\mathbf{x}))) d\mathbf{x}.$$

By dominated convergence  $G_-$  and  $G_+$  are continuous and strictly decreasing, tending to 1 as  $t \rightarrow 0$  and to 0 as  $t \rightarrow \infty$ ; for  $G_+$  this uses that  $(1 - \exp(-u))/u$  decreases strictly from 1 to 0 on  $(0, \infty)$ . Hence  $t_-$  and  $t_+$  are well defined, and  $G(t) \rightarrow 1$  as  $t \rightarrow 0$  and  $G(t) \rightarrow 0$  as  $t \rightarrow \infty$ .

The function  $\kappa$  is continuous on  $(0, \infty)$ , because  $\lambda \mapsto \lambda \kappa(\lambda)$  is Lipschitz (Mitsche & Penrose 2021, Proposition 2.3(ii) and Section 3.6), and dominated convergence then makes  $G$  continuous. To see that  $G$  is strictly decreasing, fix  $\tau > 0$ , take one realisation of  $\mathcal{P}_\tau \cup \{\mathbf{0}\}$ , and let  $\mathcal{C}(s)$  be the component of the origin when points are joined at distance  $s$ , so that  $\kappa(\tau s^m) = \mathbb{E}[1/|\mathcal{C}(s)|]$ . Components only grow with  $s$ , and when  $s_1 < s_2$  and the nearest point to the origin lies at a distance in  $(s_1, s_2]$ , the origin is alone at scale  $s_1$  and not at scale  $s_2$ , so

$$\kappa(\tau s_1^m) - \kappa(\tau s_2^m) \geq \frac{1}{2} (\exp(-b_m \tau s_1^m) - \exp(-b_m \tau s_2^m)) > 0.$$

Hence  $G$  is strictly decreasing, because its integrand decreases strictly in  $t$  wherever  $f > 0$ , and (S1) has a unique solution  $q_\gamma(f) \in (0, \infty)$ . Since  $G_-(q_\gamma(f)) \leq G(q_\gamma(f)) = 1 - \gamma = G_-(t_-)$  and  $G_-$  is decreasing,  $q_\gamma(f) \geq t_-$ , and the same argument with  $G_+$  gives  $q_\gamma(f) \leq t_+$ .

*Step 3: the quantile.* Write  $q = q_\gamma(f)$  and fix  $\varepsilon \in (0, q)$ , so that  $G(q - \varepsilon) > 1 - \gamma > G(q + \varepsilon)$ . Since  $h_n \leq r$  exactly when  $C_n(r) \leq n - k_n$ , and  $\gamma(n - 1) \leq k_n < \gamma(n - 1) + 1$ ,

$$\mathbb{P}(n^{1/m} h_n \leq q - \varepsilon) \leq \mathbb{P}(\Xi_n(q - \varepsilon) \leq 1 - \gamma + \gamma/n) \quad \text{and} \quad \mathbb{P}(n^{1/m} h_n > q + \varepsilon) \leq \mathbb{P}(\Xi_n(q + \varepsilon) > 1 - \gamma - 1/n).$$

Both tend to zero by Step 1, so  $n^{1/m}h_n \rightarrow q$  in probability.  $\square$

### S3 Proof of Counterexample 1

Let  $f(\mathbf{x}) = c_m/(1 + \|\mathbf{x}\|^{m+1})$  on  $\mathbb{R}^m$  with  $m \geq 1$ , let  $\mathbf{Y}$  have density  $f$ , and write  $a_m = c_m s_{m-1}$ . We use two elementary bounds. First,  $\|\mathbf{Y}\|$  has density  $a_m r^{m-1}/(1 + r^{m+1})$ , which for  $r \geq 1$  lies between  $a_m/(2r^2)$  and  $a_m/r^2$ , so

$$\frac{a_m}{2R} \leq p(R) := \mathbb{P}(\|\mathbf{Y}\| \geq R) \leq \frac{a_m}{R}, \quad R \geq 1. \quad (\text{S2})$$

Second,  $f(\mathbf{z}) \leq c_m \|\mathbf{z}\|^{-(m+1)}$ , so for every  $\mathbf{x}$  and every  $s \in (0, \|\mathbf{x}\|)$ ,

$$\mathbb{P}(\|\mathbf{Y} - \mathbf{x}\| \leq s) = \int_{B(\mathbf{x}, s)} f(\mathbf{z}) d\mathbf{z} \leq \frac{b_m c_m s^m}{(\|\mathbf{x}\| - s)^{m+1}}. \quad (\text{S3})$$

*Proof of Counterexample 1.* Put  $R_n = a_m n^{3/4}$ ,  $s_n = n^{1/8}$  and  $K_n = n - \omega_n = \lfloor n^{1/4}/4 \rfloor$ . Let  $N_n$  be the number of observations with  $\|\mathbf{y}_i\| \geq R_n$ , and let  $Y_n$  be the number of pairs  $\{i, j\}$  with  $\|\mathbf{y}_i\| \geq R_n - s_n$ ,  $\|\mathbf{y}_j\| \geq R_n - s_n$  and  $\|\mathbf{y}_i - \mathbf{y}_j\| \leq s_n$ . Throughout,  $n$  is large enough that  $R_n - 2s_n \geq \max(1, R_n/2)$ .

*Step 1: the number of distant observations.*  $N_n$  is binomial with  $n$  trials and success probability  $p(R_n)$ , which by (S2) lies in  $[n^{-3/4}/2, n^{-3/4}]$ . Its mean  $\mu_n$  therefore satisfies  $n^{1/4}/2 \leq \mu_n \leq n^{1/4}$ , its variance is at most  $\mu_n$ , and  $K_n \leq \mu_n/2$ . By Chebyshev's inequality,

$$\mathbb{P}(N_n \leq K_n) \leq \mathbb{P}(|N_n - \mu_n| \geq \mu_n/2) \leq \frac{4}{\mu_n} \leq 8n^{-1/4} \rightarrow 0.$$

*Step 2: no close pairs among distant observations.* Conditioning on the position of the first point of a pair and applying (S2) and (S3),

$$\mathbb{E}[Y_n] \leq \binom{n}{2} p(R_n - s_n) \sup_{\|\mathbf{x}\| \geq R_n - s_n} \mathbb{P}(\|\mathbf{Y} - \mathbf{x}\| \leq s_n) \leq \frac{n^2}{2} \cdot \frac{a_m}{R_n - s_n} \cdot \frac{b_m c_m s_n^m}{(R_n - 2s_n)^{m+1}} \leq \frac{2^{m+1} a_m b_m c_m n^2 s_n^m}{R_n^{m+2}}.$$

The right side is  $C_m n^{2+m/8-3(m+2)/4} = C_m n^{1/2-5m/8}$  with  $C_m$  depending only on  $m$ , and tends to zero for every  $m \geq 1$ . Hence  $\mathbb{P}(Y_n \geq 1) \leq \mathbb{E}[Y_n] \rightarrow 0$ .

*Step 3: conclusion.* Suppose  $N_n > K_n$  and  $Y_n = 0$ . Any observation within distance  $s_n$  of an observation  $\mathbf{y}_i$  with  $\|\mathbf{y}_i\| \geq R_n$  has norm at least  $R_n - s_n$ , so the pair would be counted in  $Y_n$ , and  $Y_n = 0$ . Each of the  $N_n$  observations beyond  $R_n$  is therefore a component on its own when observations are joined at distance  $s_n$ , so  $C_n(s_n) \geq N_n > K_n = n - \omega_n$ . Exactly  $n - C_n(s_n)$  edges of  $T$  have length at most  $s_n$ , which is fewer than  $\omega_n$ , so  $e_{(\omega_n)} > s_n$ . Therefore

$$\mathbb{P}(e_{(\omega_n)} > n^{1/8}) \geq 1 - \mathbb{P}(N_n \leq K_n) - \mathbb{P}(Y_n \geq 1) \rightarrow 1. \quad \square$$

## S4 Additional experiments

This section reports two further comparisons of the original and modified versions of lookout, and a set of showcase examples. Both versions are run as described in Section 4 of the main paper, with the default parameters  $\alpha = 0.01$ ,  $\beta = 0.9$  and  $\gamma = 0.98$ , and with scaling applied.

### S4.1 Experiment S1: Different anomaly means for Normal distribution

The non-anomalous observations have independent  $\mathcal{N}(0, 1)$  coordinates in  $\mathbb{R}^2$ , and the anomalies independent  $\mathcal{N}(\mu, 1)$  coordinates with  $\mu \in \{2.50, 2.75, \dots, 4.00\}$ . Each replication has 1000 non-anomalies and 10 anomalies, and there are 20 replications for each value of  $\mu$ . [Fig. S1](#) shows the data and results. As  $\mu$  increases the anomalies move away from the non-anomalies and the true positive rate rises. The two versions are indistinguishable at every value of  $\mu$ .

### S4.2 Experiment S2: Anomalies on the boundary as $n$ increases, Gamma distribution

This experiment repeats Experiment 2 of the main paper with  $Y_{ij} \sim \text{Gamma}(2, 2)$  for the non-anomalous points. To obtain anomalies, we generate  $n$  further points  $Y_i = (Y_{i1}, Y_{i2})$  with independent  $\text{Gamma}(2.2, 2)$  coordinates, keep those with  $\|Y_i\| > q_{0.99}$ , where  $\|Y_i\| = \sqrt{Y_{i1}^2 + Y_{i2}^2}$  is

the distance from the origin and  $q_{0.99}$  is the 0.99 quantile of the distances from the origin of the  $n$  non-anomalous points, and sample  $0.005 \times n$  of them. Each  $n$  is repeated 10 times, and the kernel density estimates use the truncated sum with 100 neighbours. Fig. S2 shows the data for  $n = 10000$  and the results. The two versions differ little; the original is slightly better at the largest sample sizes.

### S4.3 Showcase examples

Fig. S3 shows some examples of new and old versions of lookout with identified anomalies in red. The purpose of this exercise is to check if the new version agrees with the old version on obvious

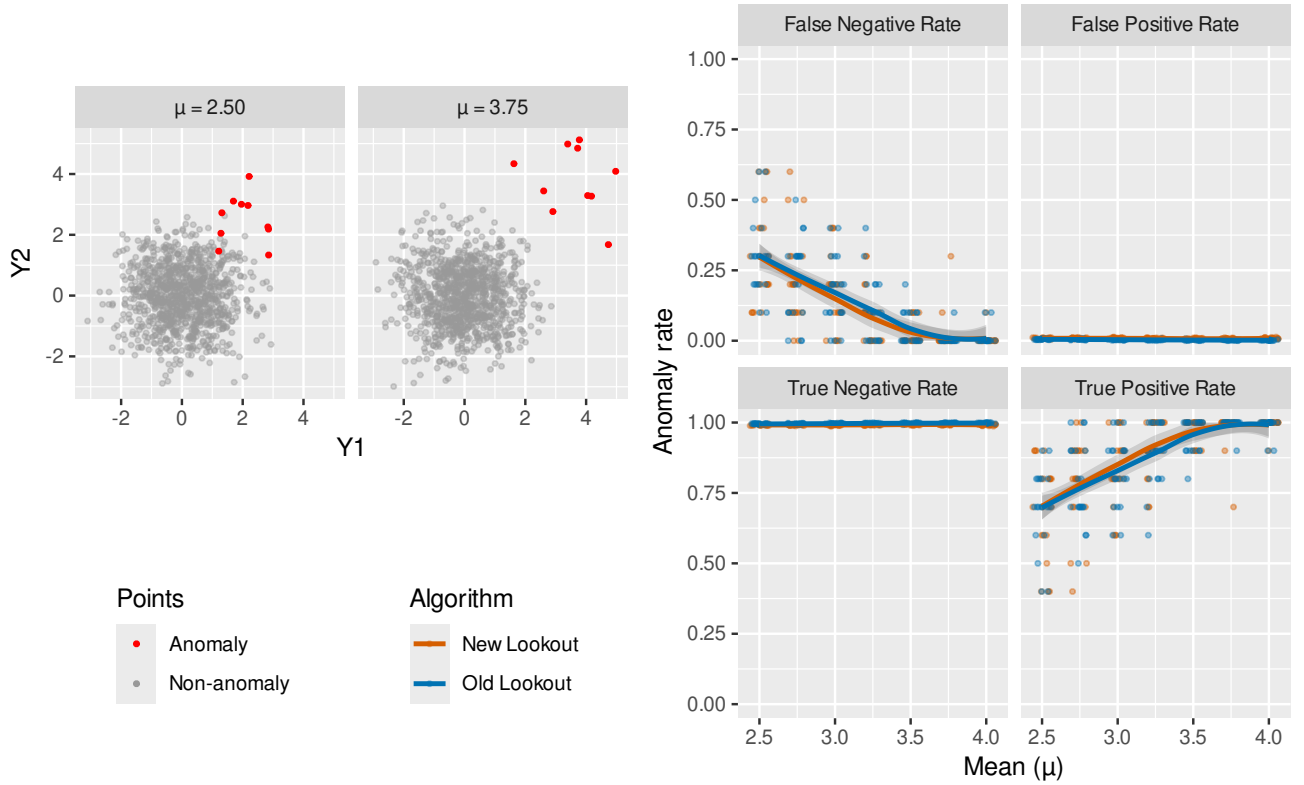


Figure S1: Experiment S1 data and results. Left: data for two iterations of the experiment with  $\mu = 2.5$  and  $\mu = 3.75$ . Anomalies with higher values of  $\mu$  are easier to identify. Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.



Figure S2: Experiment S2 data and results using Gamma distributions. Left: data for  $n = 10000$ . Right: the relative performance of the two algorithms. The points in the right panel are slightly jittered horizontally to avoid overlapping points.

anomalies. The graphs on the first row have anomalies at the top right-hand corner, which are identified by both algorithms; both algorithms find some false anomalies in the lower cloud of points, with the new version finding more. The second row depicts a bi-modal distribution with anomalies placed in the middle. Both versions of lookout identify the central anomalies, along with some false anomalies, with the new version finding more false anomalies. The third row has three clusters of normally distributed points with anomalies placed away from the three clusters. Here, the older version of the algorithm finds four false positives compared to ten with the new algorithm. The same setting is repeated in the fourth row with the anomalies spread out, and the two algorithms are similar. The last row has non-anomalous points in a parabola and the anomalous points outside of it. This is a tricky example as the points are on a manifold, and



Figure S3: Comparing new lookout with the older version on some showcase examples

we are not considering their geometry. In this example, only the new algorithm finds all three anomalies lying off the manifold, while both algorithms produce several false positives on the manifold. Overall, on these examples, the new algorithm is more sensitive to identifying anomalies, but also produces a few more false positives. The false anomalies identified by the new algorithm tend to be in regions of the data space with a low density of points, which is consistent with our definition of an anomaly.

## References

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